

PROBLEM BASED LEARNING

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Introduction

Problem based learning is a process of acquiring and understanding knowledge , skills in the context of an unfamiliar situation and applying that learning to the situation.

Definition

Problem based learning is the basic human learning process that allowed primitive man to survive in his environment.

-Barrow's and Tamblyn(1980)

Goals of problem based learning

- ✓ Construct an extensive and flexible knowledge base.
- ✓ Faster increased retention of knowledge.
- ✓ Develop effective problem-solving skills.
- ✓ Develop self-direction , lifelong learning skills.
- ✓ Become effective collaborators.
- ✓ Strengthen student's intrinsic motivation to learn.
- ✓ Develop an ability to identify relevant health problems.

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- ✓ Acquire the knowledge base necessary to define the health problem of the patients.
- ✓ Reinforce the development of effective clinical reasoning process.
- ✓ Recognize the development of effective clinical reasoning process.

Characteristics

- ✓ In this students work in small groups to increase knowledge by identifying learning objectives, engaging in self-directed work or participating in discussion.
- ✓ It provides students with greater access to information, support, resources, flexible approaches to learning, collaborative learning activities and opportunities for self-development so that can get results in higher levels of structural environment.

Principles

- ✓ The student is the focus of the educational program, the curriculum or the curriculum contents.
- ✓ The development of his/her learning capacities is emphasized.
- ✓ The problem presented in the curriculum trigger the student's abilities to analyze to understand and to solve.

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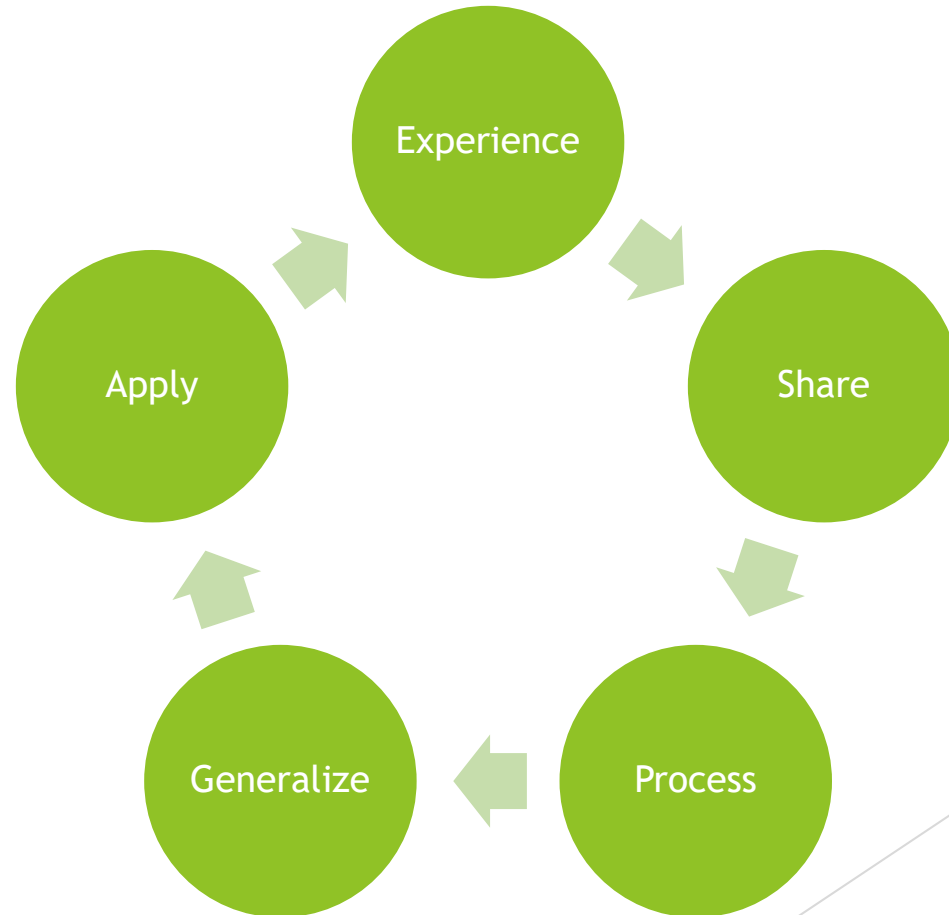
- ✓ Co-operation with other and the importance of communication is emphasized.
- ✓ Much attention is paid to the development of analytical and creative thinking skills.
- ✓ The development of self-directed learning ability.

Components

- ✓ Non-lecture format and teacher acting as a facilitator.
- ✓ Presentation of real world situation or problem from ordinary life.
- ✓ Group work and group discussion.
- ✓ Student-directed solution of the selected problem.

The Problem Based Learning Cycle

EXPERIMENTAL LEARNING



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- I. The problem based learning cycle with applying (PLANNING) EFFECTIVE USE OF LEARNING.
- II. The second step is EXPERIENCING (activity phase).
- III. The third step involves SHARING (exchanging reactions and observations).
- IV. The fourth step relates/includes PROCESSING (discussing patterns and dynamics).
- V. The fifth step related to GENERALIZING (developing and using real world principles).

Critical Thinking

It is the disciplined evidence-based process of analyzing information, making sound clinical judgements and solving problems to provide quality patient care.

Components :

Analysis:

Breaking down complex information to understand it better.

Evaluation:

Judging the reliability and relevance of information.

Inference:

Drawing conclusions based on evidence & reasoning.

Explanation:

Clearly articulating the rationale behind a decision.

Self-regulation:

Reflecting on and adjusting one's own thought process.

Problem Solving:

Applying knowledge and experience to identify and solve issues.

Clinical Judgement:

Prioritizing patient needs, interpreting data and anticipating complications.

Patient Safety:

Communication and Collaboration:

Decision making

Decision making is a crucial competency involving a combination of theoretical knowledge, critical thinking, practical experience to achieve educational and patient care goals.

It includes both analytical and intuitive processes and is taught through methods like problem-based learning and simulation to help students develop sound judgement and ethical awareness for complex clinical situations.

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Factors like experience, confidence, and the learning environment also influence a student's decision-making ability.

Problem solving Techniques

It is a systemic process that includes identifying a problem, setting goals, developing and implementing solutions and evaluating the outcomes.

Steps of problem-solving Technique :

- Identify & define the problem.
- Set goal and plan solutions.
- Implement the best solution.
- Evaluate the result.

Seven steps to problem based learning

Clarify:

The students read through the problem then identify and clarify any words or concepts that they do not understand.

Define:

The students work together to define what they think the problem is.

Analyze:

The students discuss the problem. At this stage there is no shifting of ideas.

Review:

Students now try to arrange their ideas and explanations into tentative solutions.

Identify learning objectives:

The group reaches a consensus on learning objectives.

Self Study:

Students individually gather information towards the learning objectives and prepare to share their findings with the group.

Report :

The students come together in their groups and share their results. The facilitator checks that the learning objectives have been met.

Student's Role

- Students have to assume a high degree of responsibility for their own education through effective self-learning, working with others and setting relevant goals for themselves and the group as a whole.
- Students be able to present demonstration of their learning achievements.

Group members Role

- Active participation
- Active listening
- Active asking questions
- Answering
- Giving information
- Make decisions

Common difficulties in problem based learning in group

- Unprepared for the tutorial
- Not given enough time by other members
- The dominating group members
- The group that keep storming

Tutor's/Facilitator's Role

Tutor plays a crucial role in helping the group to establish itself, setting norms for the group function, ensuring group trust, attending to the group dynamics and unique characteristics of the group.

Uses of problem based learning

- To acquire subject matter knowledge

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- Motivate learners to learn
- Help learners with retention
- Develop student's thinking skills
- Developing student's key skills relevant to employment such as interpersonal communication skills
- Fostering professional competencies and confidence together with professional learning.
- Improving the inter-disciplinary team process graduates will use in work and research.

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- Linking theory and practice.
- Encourage learners to integrate knowledge from different subjects, disciplines and sources.
- Having a sense of belonging and friendship.
- Having a sense of fun while learning.
- Expressing in operational form a philosophy of learning that is student-centric and problem-focused.

Advantages

- Problem based learning method is active and cooperative learning, the ability to think critically and clinical reasoning.
- It stimulates the students to use skills of inquiry and critical thinking, peer teaching and peer evaluation.
- It increases ability to apply knowledge in clinical situations.

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- It increases student responsibility for self-directed peer learning.
- It helps in developing flexible knowledge that can be applied to different contexts.
- This learning method helps in developing lifelong learning skills.
- It encourages students to work in teams or groups, hence by facilitating group dynamics.

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- Development of effective self-directed learning skills and increased students activity participation is facilitated.
- Increased motivation for learning is the added advantage.
- Promote collaborative learning.
- Moves learning from a passive activity to an active activity, learning becomes the act of discovery.

Disadvantages

- It is very difficult and expensive to use as a teaching technique when the class size is large.
- Students require orientation to perform the role of a learner in PBL setting.
- Evaluation is quite difficult and sometimes may be subjective.

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- Resource expensive.
- Staff and students may be initially uncomfortable with problem based thinking because they are used to subject based learning and they do not really understand how to proceed in problem based learning.
- Measurement of learning outcomes is difficult.

Conclusion

Problem based learning is an effective educational approach that transforms nursing students from passive learners into active problem solvers. By promoting critical thinking, collaboration and real world application, problem based learning helps produce competent, confident and reflective nursing professionals which is the ultimate goal of nursing education as emphasized by Bhaskar NIMA and Bhaskar Elakkuvana.

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